Abhishek Dass

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SUMMARY

Computer Science and Business Systems graduate with hands-on experience in flight software, embedded systems, wireless telemetry, real-time data acquisition, and command systems. Member of the Estonia Student Satellite flight software team, contributing to Core Flight Software (cFS) applications for the satellite command module. Previously led Data Acquisition for a Formula Bharat electric vehicle, developing real-time telemetry and sensor-data systems. Interested in wireless communication, RIS/6G systems, satellite communications, and research-driven engineering.

TECHNICAL SKILLS

Programming: C, C++, Python, SQL

Embedded & Flight Software: Core Flight Software (cFS), Arduino, ESP32, STM32, RTOS, sensor integration, real-time systems

Wireless & Communication: RF communication, telemetry, CAN, UART, SPI, IZC, MQTT, communication protocols

Data & Instrumentation: LabVIEW, data acquisition, real-time telemetry, data logging, ETL pipelines

Engineering & Tools: Git, GitHub, MATLAB, SolidWorks, Fusion 360, Linux

Concepts: Computer Networks, Embedded Systems, OOP, Data Structures, System Design, Machine Learning

EXPERIENCE

Flight Software Team Member 2025 – Present

Estonia Student Satellite Program

- Contribute to Core Flight Software (cFS) applications supporting the satellite's command module and flight-software architecture.
- Work on embedded software and system-level problem solving for a research-oriented satellite platform.
- Gain practical exposure to command handling, telemetry, real-time software, and reliable subsystem communication.

Aerodynamics & Data Acquisition Lead Mar 2023 – Jan 2026

E-Formula AshwaRiders

- Led sensor integration and real-time data acquisition for a Formula Bharat electric racing vehicle.
- Designed telemetry pipelines capturing speed, thermal, battery, and vehicle-performance data from embedded sensors.
- Developed LabVIEW-based monitoring and data-processing systems for real-time engineering analysis.
- Worked with embedded hardware, wireless communication, CAN and sensor interfaces under real-time reliability constraints.

Junior Software Developer Jul 2025 – Jan 2026

PugArch Private Limited

- Developed Python-based data-processing and ETL pipelines handling 10,000+ structured records.
- Integrated database systems and automated preprocessing workflows, reducing manual processing effort.
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PROJECTS

Real-Time Vehicle Telemetry & DAQ System | Python | C/C++ | LabVIEW | PostgreSQL | MQTT | Embedded Systems

- Built an end-to-end telemetry pipeline from sensor acquisition and packet transmission to parsing, processing, visualization and data logging.
- Designed structured communication and data-processing workflows for continuous vehicle telemetry.
- Implemented real-time monitoring of vehicle and battery parameters for engineering analysis.

RIS-Assisted LEO Satellite Communication | Python / MATLAB | Wireless Communication | RIS | Optimization

- Developing a simulation framework to study Reconfigurable Intelligent Surfaces (RIS) for satellite-to-ground wireless communication.
- Model satellite-ground channels and RIS phase configurations to evaluate received power, SNR, BER and communication performance.
- Exploring dynamic RIS configuration and optimization for changing LEO satellite communication conditions.

EDUCATION

St. Vincent Pallotti College of Engineering and Technology, Nagpur

B.Tech — Computer Science and Business Systems